

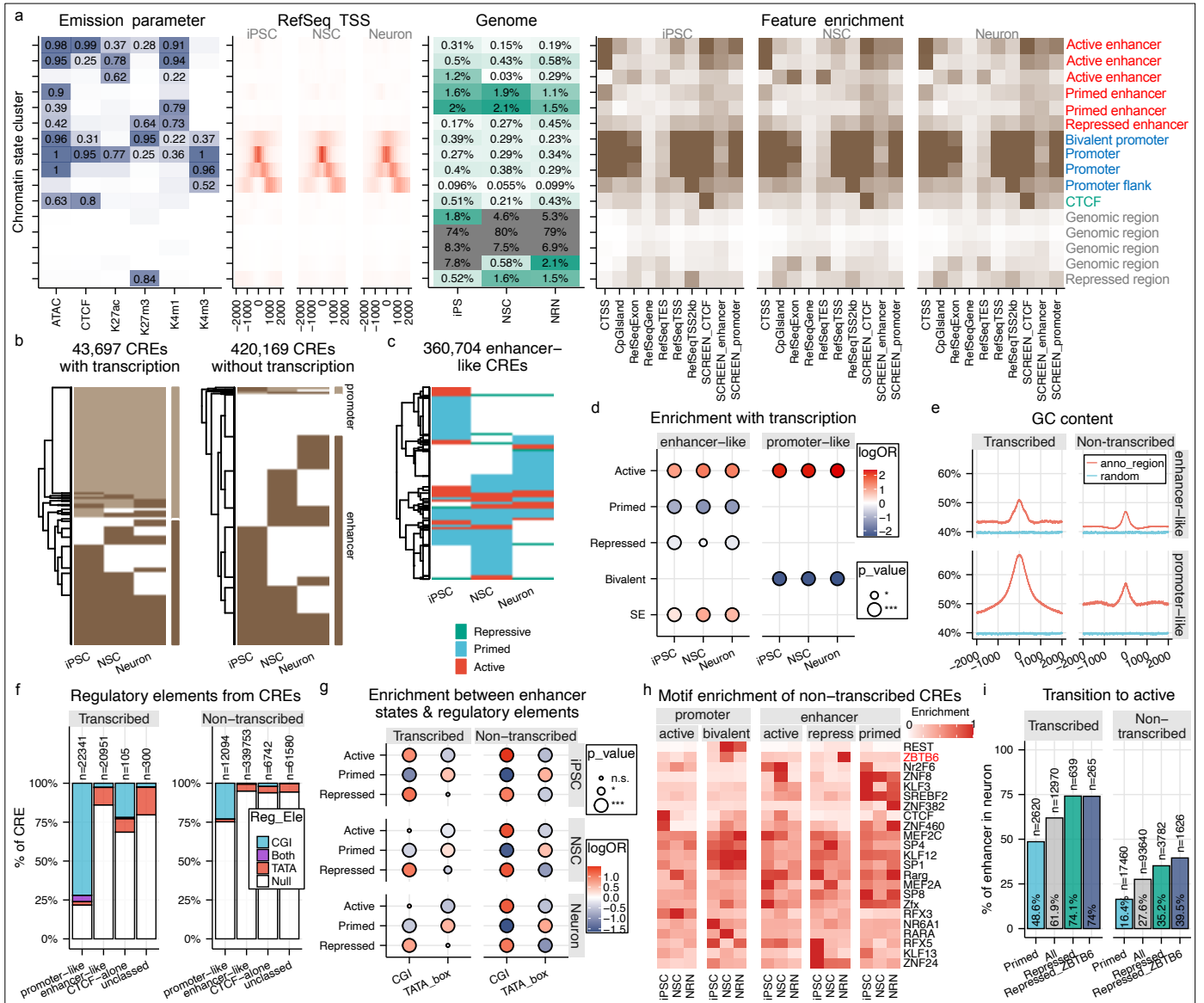


Figure 2 | Chromatin states of CREs correlate with transcription output across differentiation.

a, Genome-wide chromatin state clustering of iPSC, NSC and Neuron determined by ChromHMM using scATAC-seq (grouped by sample) and CUT&Tag data on H3K4m1, H3K4m3, H3K27ac, H3K27m3 and CTCF. Chromatin states were manually curated based on emission profiles. **b**, Heatmaps illustrating the promoter-typing of the CREs, grouped by the presence or absence of TSS signal across the three samples. **c**, Heatmap of all enhancer-like CREs categorized by chromatin states (repressive, primed and active) across differentiation. **d**, Enrichment between transcription states and chromatin states of enhancer-like and promoter-like CREs. **e**, GC content of CREs grouped as transcribed and non-transcribed, and promoter-like and enhancer-like. The x-axis represents the distance (nt) from the CRE mid-point. **f**, Percentage of CREs containing specific regulatory elements, including CpG islands (CGI) and TATA boxes. **g**, Enrichment between regulatory elements (CGI and TATA box) and chromatin states of enhancer-like CREs. **h**, Motif enrichment analysis on CREs with different chromatin state highlighting significant motifs (ENR_RATIO > 3 & q-value < 0.01 from 400 TF motifs with expressing TFs) from non-active groups. **i**, Number and percentage of neural enhancer-like CREs with active chromatin state in at least one of the three samples across differentiation. Fisher's exact test was used in (**d**) and (**g**) to calculate odds ratios (OR). $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.